

Tennis Quant Analyst Test

Predictive Modeling and Forecasting Analysis

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Relevant functions are presented in python files named according to sections (e.g, section_1.py etc).

1. The Data

1.1. Participant Ordering Bias

Question: Briefly explain why it may be problematic, from a modelling perspective, that participant1 is always the winner of the match. Then reorder participants 1 and 2 alphabetically by player name.

Answer:

In models that rely on feature engineering such as traditional classification machine learning approaches (XGBoost, Random Forests, Logistic Regression etc.), explicitly coding player 1 as the winner every time introduces bias/data leakage into the model. If individual player IDs are input (as would be expected for identifiability), the model can simply learn to predict ‘player_1_id’ and achieve 100% accuracy irrespective of additional features, leaving the model with no useable signal and thus not learning what distinguishes a winner from a loser.

In the case of the Bradley-Terry model presented in part 3, this should technically matter less, as player-specific coefficients are input into the model instead of explicit column positions (e.g., Alcaraz could be identified as player_1 in one match, and player_2 in another; this model just ‘sees’ Alcaraz’s specific coefficient). However, the beta intercept could ‘learn’ a systematic weighting and lose interpretability (let’s not get ahead of ourselves, however, we’ll discuss this in 3.1).

Alphabetical swapping has been implemented with the *reorder_participants_alphabetically* function. This followed typical conventions of utilising the surname as the metric to reorder around. A unit test is supplied alongside the function to check for any implementation errors

1.2. Home Advantage Indicator

Problem Statement: Add a column `homeadv` to the dataset. This variable should take the value 0 if either both players, or neither player, are playing in their home country; 1 if only participant 1 is playing in their home country; and -1 if only participant 2 is playing in their home country.

Answer:

The column is added to the dataframe utilising the *add_home_adv* function, with an assert that both columns are either 0 or 1 prior to calculation.

2. Descriptive Statistics

2.1. Games per Set by Surface and Round

Problem Statement: Using the data provided, investigate whether the average number of games per set depends on (i) the surface and (ii) the round.

Answer: As exploratory data analysis can sometimes be a long and messy affair, I shall try to present the workflow of exploration within this answer section, and present final visuals or analyses in the following Analysis section. I also believe there are some factors that underpin the investigation that could technically fall under the scope of question 2.2, but they are addressed here initially.

Average games per set was calculated as the total number of games in a match divided by the total number of sets.

First, it's prudent to check for valid matches. The function *set_validity_check* asserts only valid set numbers within the *_sets_won* columns, and then checks that the winner is reflected in the number of sets won. There were 271 matches where set numbers are equal or do not reflect the match winner, and 567 where there were a total of 1 set; a quick online check of 3 of these matches showed they were most likely early retirements, so whilst a valid results for match winner, they may bias future models as a win by retirement doesn't fully reflect skill. These matches were removed from the dataset.

Next, we must consider that tiebreak games may influence our comparisons. A tiebreak game would typically be indicated in any set where there is a total of 13 games (7-6 or 6-7). Historically, some tournaments would also use a no-tiebreak deciding set, where the number of games played could theoretically be infinite. Inspection of initial distribution plots split by surface type (Appendix 1) shows some large outliers at average number of games higher than 13, with a maximum of 23.7. Whilst the bulk of observations - regardless of surface type - look to be grouped together, outliers are likely to compound mean comparisons or statistics comparisons (should these be warranted). To overcome this, these outliers can be trimmed to a theoretical maximum of 13. Whilst an argument could be made to totally remove these games, they still represent valid and important signal for the players involved within the game. Trimming the maximum average games per set to 13 (the standard tiebreak score) still maintains the players' strong win signal, but doesn't corrupt underlying signals. There is an important caveat that we're trimming the *match* level average, and therefore some no tiebreak decider sets may still be included, but given the granularity of the data is fixed at match accumulate games, it is an assumption we have to make.

Surface Type Comparing surface types on trimmed games, we keep Indoor_Hard (with a quick name change) separate from Hard due to confounding weather differences despite the surface technically being the same. Descriptive statistics and distribution plots are presented in Table 1 and Figure 1 via the *compare_surfaces* function. As aforementioned, the bulk of observations regardless

of surface type look to be grouped fairly well, with the majority of observations showing a normal distribution, with a ‘bump’ of observations concentrated around average number of games of 13 (reflecting the prevalence of 7-6 or 6-7 scores). Interpretation of mean and median values and standard deviations also show little difference, with the maximum mean difference being between Clay and Grass, with a difference of roughly 0.4 games per set. On a population level, average number of games does depend on surface type, though the practical effect is minimal. If we were to complete further statistical comparisons (say, a Kruskal-Wallis test or one-way/Welch’s ANOVA) these are likely to report statistically significant differences, due to the large sample size (and whether ‘statistically’ significant = meaningful is debatable). Instead, calculating the eta-squared (a measure of ‘amount’ of variance explained in average games per set) reports surface accounts for 1.01% of variance ($\eta^2 = 0.0101$); a small effect.

On a population level, and if outcomes are being modeled at a match winner level in lieu of any serve specific information, this may be negligible. However, the influence of surface becomes important if matches are modelled from a point-level perspective (utilising a Barnett-Clark approach, for example), or if athletes are considered on an individual basis (where some will favour a particular surface).

Table 1: Descriptive statistics of trimmed dataset by surface type

Surface	Count	Mean	Median	Std	Min	Max
Clay	12,776	9.649	9.500	1.328	6.0	13.0
Grass	5,030	10.067	10.000	1.327	6.0	13.0
Hard	16,904	9.816	9.667	1.340	6.0	13.0
Indoor_Hard	6,274	9.925	10.000	1.367	6.0	13.0

Figure 1: Distribution plots by surface type

Round Round differs slightly from surface analysis as there is an inherent temporal progression. Bronze ($n = 3$) and Round Robin ($n = 335$, a year-end championship format) were excluded as they don’t fit the standard tournament progression (and had low sample size for Bronze). These were ordered chronologically: three qualifying rounds (Q-First, Q-Second, Qualification), four early-to-mid main draw rounds (Round 1 through Round 4), and three late-stage rounds (Quarter Final, Semi Final, Final). Descriptive statistics are presented in Table 2 and visualised in Figure 2, calculated and plotted utilised the *compare_rounds* function.

As with surface, visual inspection of distribution plots, and analysis of descriptive statistics, show little difference between rounds, with the maximum difference between Q-First and Final being 0.43 games. A smaller and general increase can be observed as round increases. As with surface, on a population level (due to the high sample size) this is likely to be statistically significant, but trivial in practice. Eta-squared for round across all 10 categories is $\eta^2 = 0.0075$, meaning round

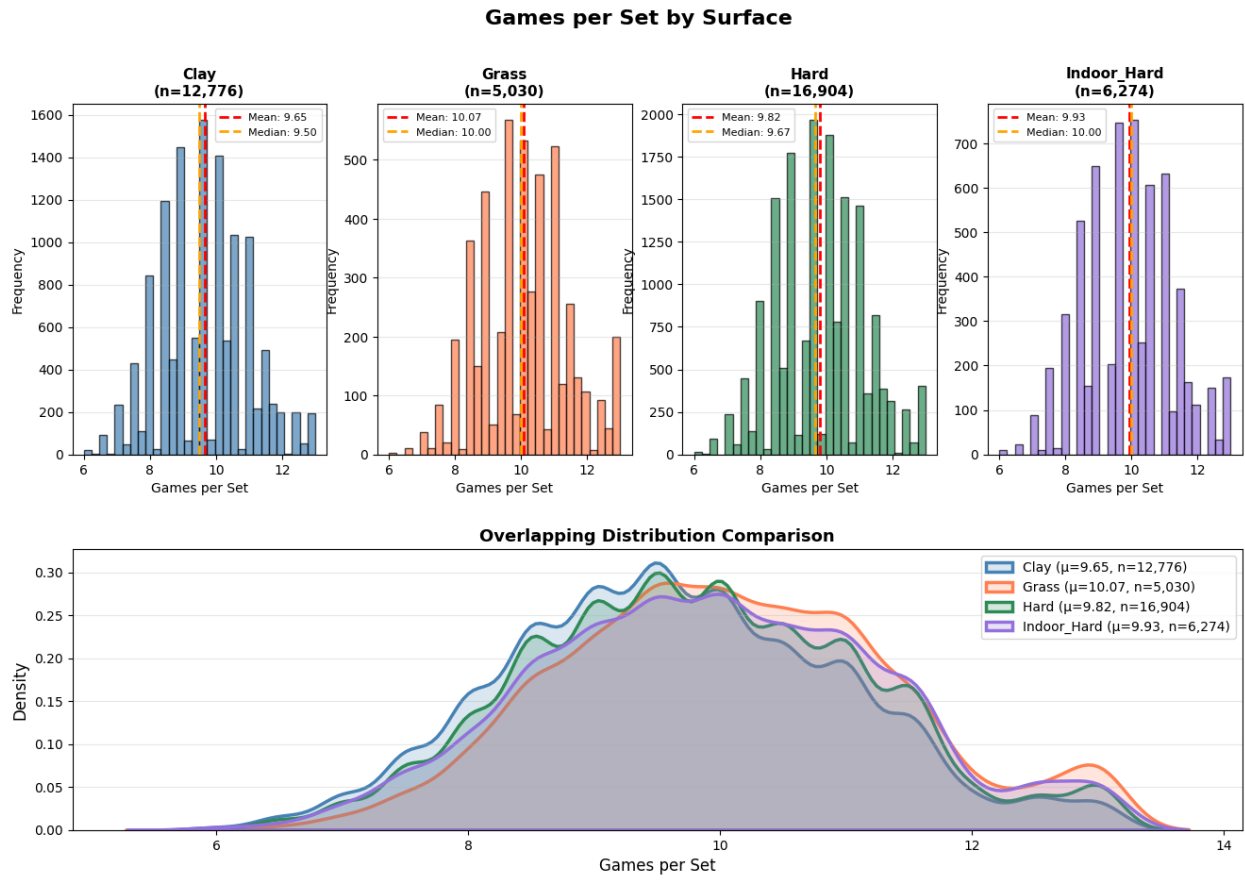


Figure 1: Games per Set by Surface

explains approximately 0.75% of the variance in games per set — even smaller than the surface effect (1.01%).

Formal trend testing within the main draw confirms a small but detectable progression: Spearman $\rho = 0.025$ ($p < 0.001$), and a linear regression of games-per-set on round number yields a slope of 0.022 games per round (95% CI [0.013, 0.032], $p < 0.001$). Across the seven main-draw rounds this amounts to a total expected difference of about 0.15 games per set — statistically significant due to sample size, but trivial in practical terms.

From a modelling perspective, neither surface nor round meaningfully changes match-level games per set. Both factors may matter more in finer-grained models — point-by-point or player-specific approaches — but for population-level summaries of match competitiveness, neither is a strong predictor.

Table 2: Descriptive statistics of trimmed dataset by round

Round	Count	Mean	Median	Std	Min	Max
Q-First	8,533	9.606	9.500	1.399	6.000	13.0
Q-Second	1,955	9.755	9.667	1.382	6.000	13.0
Qualification	3,527	9.803	9.667	1.334	6.000	13.0
Round 1	11,913	9.851	10.000	1.305	6.000	13.0
Round 2	7,649	9.891	10.000	1.333	6.000	13.0
Round 3	1,882	9.894	10.000	1.292	6.000	13.0
Round 4	537	9.895	9.750	1.289	6.500	13.0
Quarter Final	2,613	9.902	10.000	1.327	6.500	13.0
Semi Final	1,347	9.996	10.000	1.373	6.333	13.0
Final	690	10.038	10.000	1.316	7.000	13.0

Figure 2: Distribution plots by round

2.2. Other Factors Affecting Games per Set

Problem Statement: What other factors might be significant in explaining the number of games per set?

Answer:

As discussed within section 2.1, match format is important to consider, as the final set rules (whether a no tiebreak deciding set is used) can drastically skew the upper quartiles of the average games per set. (As an aside, I checked all the ATP matches (which shows specific set scores) between 2014 and 2024 in the Jeff Sackman open source database, showing 107 (0.35%) of matches ended in a no tiebreak deciding set, with a maximum upwards of 50 games in the final set).

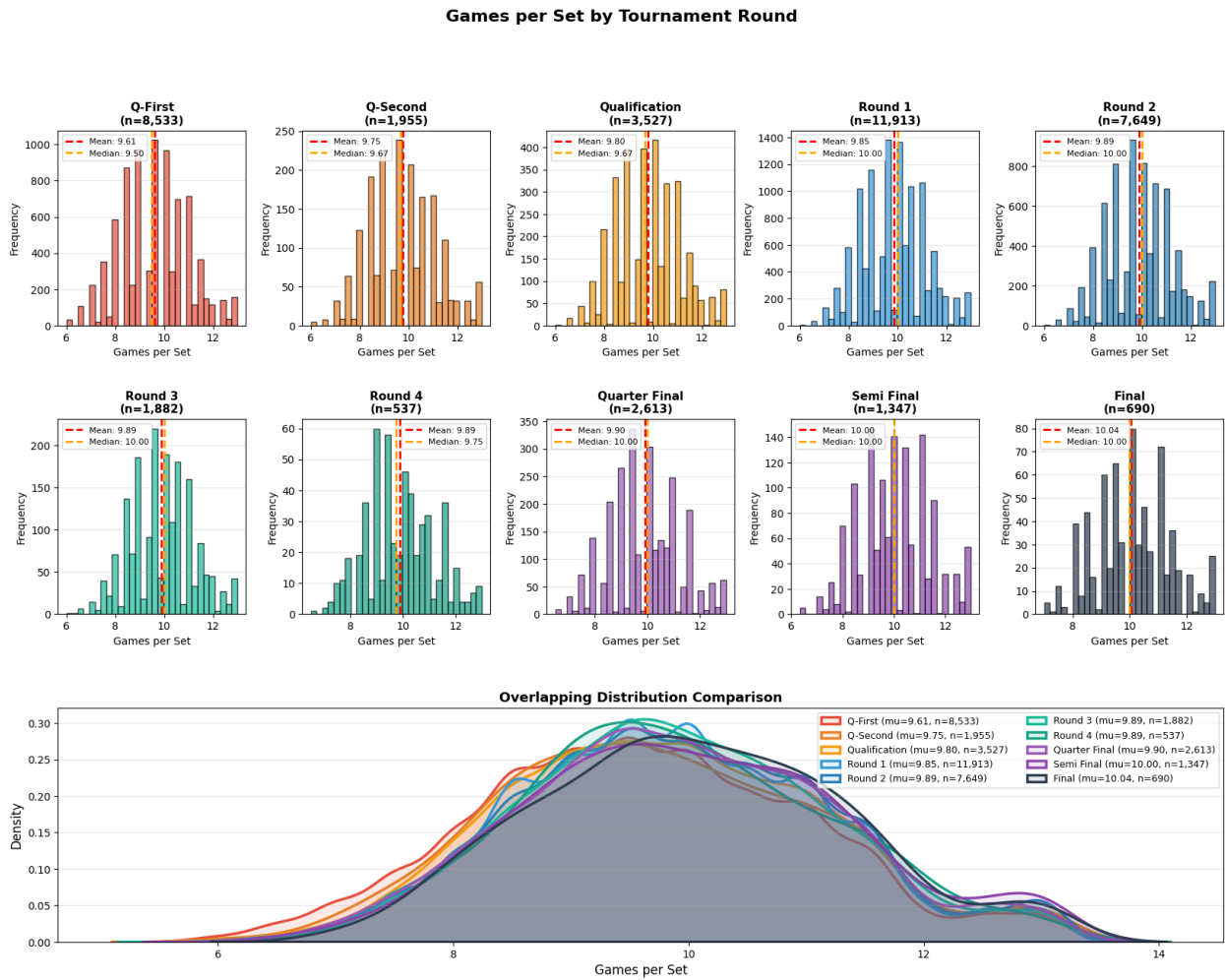


Figure 2: Games per Set by Round

Relative player level/strength is also a factor that could influence average games per set. Typically, one would expect a stronger player to beat a weaker player, with the greater the margin between these, the greater the probability of the stronger player winning. It is possible that the greater the relative player strength is, the lower the average number of games per set would be. This could easily be assessed using inherent tennis ranking systems (ATP rankings), self-calculated ELO rankings, or alternative approaches such as Bayesian rankings.

Viewing these factors in isolation would also only show part of the puzzle; interpreting the influence of variable interactions with each other also becomes important to consider when enough matches are present. For example, checking whether round influences hold similar trends across surface type.

2.3. Dataset Filtering

Problem Statement: Trim the dataset to keep only matches in which both players have played at least 10 matches in the dataset.

Answer:

The dataset is filtered using the `filter_min_matches` function, which iteratively removes players with fewer than 10 matches until the player set stabilises. A single-pass approach was insufficient, as removing under-threshold players can push their opponents below the threshold, violating the requirement that both players in any retained match have at least 10 matches in the filtered dataset. The filter converged after 4 iterations, retaining 657 players across 38,265 matches. Minimum match count per player is now 10.

Note - as a result of identifiability issues discussed in 3.1, the function also includes a iteration of evaluation of players who have either zero wins or zero losses across all their matches. A further 8 players were removed, leading to 649 players across 38,180 matches.

3. A Simple Model

3.1. Model Identifiability

Problem Statement: Do you need any additional assumptions or constraints to make this model identifiable? Should the intercept term be set to $\alpha = 0$?

Answer:

Model A presents a Bradley-Terry model. The main constraint to apply is a method by which to make player identifiable within the model - such as a sum-to-zero constraint. This enables two individual players to be input and have their relative strengths (not absolute) be utilised to produce

a valid pk. Furthermore, forcing the sum of the parameters (or their log values) to zero minimizes the sum of the variances of the estimated scores. Were we to use a different approach, such as a fixing a reference player and calculating all players scores relative to this player, all the uncertainty in the reference players score gets ‘absorbed’ by the other players. In a sum-to-zero constraint, this is distributed equally.

A potential downside of this constraint is that each player strength is evaluated relative to the average player; not an absolute or external benchmark. This means that if we were to remove a player, or trim the dataset, relative player strengths would alter (to what degree is dependant upon the trimming of the dataset). However, as we are already utilising a trimmed dataset with a minimum number of games per player, and purely within a male/single tour population, this downside should be minimal.

A second identifiability issue arises when a player has either zero wins or zero losses across all their matches in the dataset. In such cases, the maximum likelihood estimate for that player’s μ does not exist: the likelihood is monotonically improved by sending μ to $-\infty$ (for an all-loss player) or $+\infty$ (for an all-win player), and the optimizer terminates at large finite values determined only by numerical tolerance. We address this by restricting the dataset to players with at least one win and at least one loss, losing a total of 9 players.

The intercept term of beta should be set to zero under these constraints. The beta coefficient expresses or ‘soaks up’ residual variance in the model expressed as a function of column position, outside of the player coefficients. As an example, if beta were allowed to be expressed and given a value, if this value were anything outside of zero, and two identical players were input into the model, we would end up predicting one player to have an advantage by virtue of the non-zero beta (when identical players should result in a 50/50 prediction, all other variables being equal).

3.2. Model A - Fitting and Analysis

Problem Statement: Assuming match results are independent of each other, fit this model by maximum likelihood on the data provided and analyse the parameters you obtained.

Answer:

An initial Model A was fitted on 649 players over 38,180 matches. The final value of the likelihood was -23436.28. The AIC was 48168.56. The model was fit utilising the statsmodels GLM.

Evaluating the highest strength and lowest strength players elicited common sense results. The bottom 5 players have all only have one win. Although the end of the dataset is 31/12/2024, and hence the top 5 does not reflect the ATP ranking at that time, the rating system is ‘time blind’ and purely rates all players on all their career matches. The top five align with other colloquial ‘all time best’ lists (<https://www.ultimatetennisstatistics.com/peakEloRatings>).

Table 3: Top 5 and bottom 5 rated players from Model A

Category	Player	Alpha	Matches
Top 1	Novak Djokovic	3.423	643
Top 2	Roger Federer	3.158	368
Top 3	Rafael Nadal	2.865	499
Top 4	Carlos Alcaraz Garfia	2.644	257
Top 5	Jannik Sinner	2.458	327
Bottom 1	Gonzalo Escobar	-2.937	10
Bottom 2	Orlando Luz	-2.814	10
Bottom 3	Harry Bourchier	-2.716	12
Bottom 4	Michael Geerts	-2.708	13
Bottom 5	Ti Chen	-2.485	11

3.3. Model B - Adding Home Advantage

Problem Statement: Modify Model A to fit a new model Model B, which includes a home effect.

Answer:

Model B introduces a home advantage term by adding γh_k to the linear predictor, where h_k is the home indicator from 1.2. The match probability becomes

$$p_k = \frac{1}{1 + \exp(-(\alpha_{i(k)} - \alpha_{j(k)} + \gamma h_k))}.$$

The signed encoding of h_k (+1 if participant 1 is home, -1 if participant 2 is home, 0 otherwise) means a single coefficient γ captures home advantage symmetrically.

Model B elicits a likelihood values of -23436.26, barely indistinguishable from Model A. AIC marginally increased to 48170.52. Common findings within sports dictate that the home team should have an advantage (tennis literature has shown this in male games <https://doi.org/10.1080/02640414.2010.516762>). However, investigating the underlying raw data, we see that home is actually at a disadvantage within this data:

Table 4: Home advantage distribution and participant 1 win rates

Home Advantage	Count	% of Matches	P1 Win Rate	Interpretation
home_adv = -1 (P2 home)	4,467	11.7%	0.537	P1 wins 53.7% when opponent is home

Home Advantage	Count	% of Matches	P1 Win Rate	Interpretation
home_adv = 0 (neutral)	29,806	78.0%	0.499	P1 wins 49.9% in neutral matches
home_adv = 1 (P1 home)	3,907	10.2%	0.452	P1 wins 45.2% when playing at home

This could potentially reflect a selection bias - where top players who are likely to travel to further afield tournaments naturally win more (as a function of being a ‘better player’) compared to players of lower rank or may play more home tournaments. Splitting players into quintiles based on alpha scores, Table 5 (derived from *check_adv_stats* function) shows that weaker players do proportionally play more games at home, hence potentially contributing to the lack of affect within Model B.

Table 5: Travel patterns by player strength quintile

Quintile	% Home	% Away	% Neutral	Avg matches
Q1 (weakest)	25.9	6.9	67.2	23.1
Q2	20.9	9.8	69.2	46.2
Q3	14.5	10.2	75.3	82.5
Q4	13.6	11.0	75.4	156.3
Q5 (strongest)	10.2	10.7	79.0	280.0

4. An Alternative Model

4.1. Probability Distribution for Games Won

Problem Statement: If the total number of games played in the match is N , what probability distribution might be a natural starting point for modelling Y conditional on N ?

Answer:

The natural starting point for modelling the distribution of Y_k conditional on N_k would be the Binomial distribution. This treats each of the N_k games within match k as an independent Bernoulli trial, where player $i(k)$ wins each game with probability p_k :

$$Y_k \mid N_k \sim \text{Binomial}(N_k, p_k)$$

where p_k retains the Bradley-Terry parameterisation from Model A.

This naturally generalises Model A: setting $N_k = 1$ recovers the Bernoulli match-level model exactly (working under an IID assumption), but the Binomial form allows the model to use richer game-level information. For example, a 6-0, 6-0 result and a 7-6, 7-6 result both count as a win in Model A, but both represent very different underlying information about the players input into the model, offering increase granularity.

The likelihood contribution for a single match k is:

$$L_k(\alpha) = \binom{N_k}{Y_k} p_k^{Y_k} (1 - p_k)^{N_k - Y_k}$$

Taking logs, the contribution to the log-likelihood is:

$$\ell_k(\alpha) = \log \binom{N_k}{Y_k} + Y_k \log p_k + (N_k - Y_k) \log(1 - p_k)$$

The binomial coefficient $\binom{N_k}{Y_k}$ depends only on the data, not on α , so it acts as an additive constant in the log-likelihood and does not affect the maximum likelihood estimate.

As mentioned above, this model assumes games within a match are conditionally independent given player strengths, which is a simplification — in practice, within game factors can alter this IID assumption, such as server/returner interaction, player momentum, and pressure of specific points. However, it is a useful and information place to start at a whole level.

4.2. Modelling Game Win Probability

Problem Statement: How could you model p using player strength parameters and ?
Write down the full log-likelihood for matches $k = 1, \dots, K$.

Answer:

The natural choice is to retain the same Bradley-Terry parameterisation used in Model A, but applied at the game level rather than the match level:

$$p_k = \frac{1}{1 + e^{\alpha_{j(k)} - \alpha_{i(k)}}}$$

The interpretation carries over directly from Model A. p_k now represents the probability that player $i(k)$ wins any individual game against player $j(k)$, dictated by the relative strength difference $\alpha_{i(k)} - \alpha_{j(k)}$. The same identifiability constraint discussed within 3.1 still applies — an additive shift across all α values leaves every p_k unchanged — so we maintain the sum-to-zero constraint, and again set $\beta = 0$ for the same reasoning given in 3.1.

Assuming matches are independent of one another, the joint likelihood across all K matches factorises into the product of per-match Binomial contributions from 4.1:

$$L(\alpha) = \prod_{k=1}^K \binom{N_k}{Y_k} p_k^{Y_k} (1 - p_k)^{N_k - Y_k}$$

Taking logs converts this product into a sum, yielding the full log-likelihood:

$$\ell(\alpha) = \sum_{k=1}^K \left[\log \binom{N_k}{Y_k} + Y_k \log p_k + (N_k - Y_k) \log(1 - p_k) \right]$$

As noted in 4.1, the binomial coefficient $\binom{N_k}{Y_k}$ depends only on the observed data and is therefore constant with respect to α . Substituting the Bradley-Terry form for p_k and dropping the constant gives the working form:

$$\ell(\alpha) = \sum_{k=1}^K \left[Y_k (\alpha_{i(k)} - \alpha_{j(k)}) - N_k \log(1 + e^{\alpha_{i(k)} - \alpha_{j(k)}}) \right]$$

This again mimics the identifiability constraint, where the log-likelihood depends only on strength differences between players, not on absolute values.

4.3. Fitting the Game-Level Model

Problem Statement: Fit this game-level model to the data provided. Analyse and interpret the parameter estimates you obtain.

Answer:

Model C is fit almost identically to Model A and B, although statsmodels GLM was used with the Binomial family passing the 2-column vector of [participants1_games_won, participant2_games_won]. The log-likelihood was -92597.77 and AIC 186491.54. In isolation, these figures can't really be interpreted alongside Models A and B, but the rankings presented in Table 6 (at least at the top end of the table) do reflect those of the match level models. As the players ranked at the bottom are likely to have the minimum number of games (and hence more uncertainty in their predictions) it is likely that these may not entirely mimic the whole match models.

Table 6: Top 5 and bottom 5 rated players from Model C (game level)

Category	Player	Alpha	Matches
Top 1	Novak Djokovic	0.706	643
Top 2	Rafael Nadal	0.664	499
Top 3	Roger Federer	0.625	368
Top 4	Carlos Alcaraz Garfia	0.552	257
Top 5	Jannik Sinner	0.512	327
Bottom 1	Luis Patino	-0.597	10
Bottom 2	Sasi Kumar Mukund	-0.585	13
Bottom 3	Ivan Nedelko	-0.525	15
Bottom 4	Ryan Shane	-0.523	10
Bottom 5	Daniel Nguyen	-0.519	10

4.4. Model Assumptions

Problem Statement: What assumptions are implicit in this model? Which of them may be unrealistic in the context of tennis?

Answer:

The majority of assumptions (unrealistic or not) within this model can be grouped under an assumption of ‘constant-ness’.

A primary assumption already discussed in section 4.1 is the IID assumption, where each game is assumed to be independent of each other and hence p_k is kept constant. This most likely isn’t valid on two fronts. Firstly, the influence of momentum and within game/set dynamics are likely to influence the probability of winning further games. Secondly, serving as a significantly structural advantage (averaging around 63% within ATP but this does vary by surface). This serve can then also interact with different players returning strength (hence in inclusion of both in popular models like the Barnett-Clarke point based model). At a whole match level, it is possible that these influences can get summarised or ‘folded into’ a holistic win probability, but if specific sets are being modelled, it could be important to note who served first in the very first game.

A secondary assumption is that player strength is kept constant over time (or at least, weighted towards whatever year the player played the most matches). It is extremely likely that a player’s strength - and probably playing dynamics - are going to vary throughout their career, and hence this assumption is quite unrealistic.

A third assumption is that player strength is kept constant regardless of surface type. Although EDA within section 2 did show that, on a population level, surface may only have a small effect, at an individual level player surface preference is well established. A single player strength cannot capture this, and may over or underestimate player winning probabilities in certain situations.

4.5. Predicting Match Outcomes from Game-Level Model

Problem Statement: How could this game-level model be used to predict match outcomes? Describe possible approaches and rank them.

Answer:

At its simplest, given the natural generalisation of Model A to this current model, the model could be directly used to simply ‘pick a winner’. If player i is ‘stronger’ than player j at a game level, and hence are more likely to win, they are more likely to win over the match level; a valid assumption under the Binomial model. However, output probabilities, or our confidence in the prediction, will not align (and will be somewhat dependent upon the length of match; 3 or 5 sets) given the way tennis’s recursive set structure amplifies small per-game advantages into larger per-match ones.

A secondary modelling option could be to include a surface coefficient within the Binomial model. On a simple level, this could simply be an additional surface term and hence a uniform coefficient for each surface (in line with previous findings, however, this probably would not achieve much increase in predictive power). A different option could be to include a player-by-surface interaction term to formally capture how different players perform on different surfaces. This could then further be extended as above if the aim is simply to predict the winner.

For more of a different aggregation approach, either of the two previous approaches could be used to calculate the game specific p_k for each player respectively, and then build a Monte Carlo simulation logic mapping game win probabilities to set win and match win probabilities, sampling over 10,000 (or any high number really) steps based on the ‘best_of’ sets column. Whilst this theoretically should result in identical predictions for ‘who won the match’, the output probabilities for this match win should be better aligned with game level performance.

A final option, and probably outside of the scope of the current task, is to fit any of the previously discussed models under a Bayesian framework. This enables a proper posterior distribution over player strength instead of a point estimate, and allows for uncertainty propagation through the Monte Carlo simulation framework. Whilst priors will have to be decided (either use of ‘weakly informative’ priors or defined based on a training dataset), these can also be applied to surface. Whilst this added dimension of uncertainty may not influence players with a lot of games, it could be useful in better aligning predicted probabilities for low-game players.

Expected predictive performance, best to worst on probability calibration: 1. Bayesian + surface + Monte Carlo 2. Surface + Monte Carlo 3. Monte Carlo only 4. Direct α comparison. On binary winner accuracy the differences are much smaller, since most matches have a clear favourite identified by all approaches.

Under the time constraints of the current task, I shall utilise the simple Model C defined within

section 4.3, but apply outputs to a Monte Carlo simulation for match level prediction.

Monte Carlo match simulation logic is presented in *simulate_game* (*simulate_tiebreak*), *simulate_set*, and *simulate_match*. Once P_k has been computed by respective models, matches within a dataframe can be bath ran through *run_match_simulations*.

5. Forecasting

5.1. Computing Predicted Probabilities

Problem Statement: For each of Models A, B, and C, compute the predicted probability that participant 1 wins for all matches played after January 1, 2020.

Implementation:

Upon attempting to fit Model A, B and C, I found the statsmodels GLM (and general processing of these models) was too cumbersome for effective forecasting and multiple refits (for Model A, a predicted total processing time of 6 hours with weekly refits of data). After some light chat-botting, it was found that models could be fit utilising Sklearn’s LogisticRegression, with *fit_intercept* set to False and a barely-zero L2 penalty to solve the same penalised-likelihood problem (although doesn’t calculate the additional standard errors, deviance and information matrices that statsmodels performance). To verify the two implementations were equivalent for prediction, both were fith on the full filtered dataset as used for all other models.

Table 7: Comparison of statsmodels and sklearn fits on the full filtered dataset (38,180 matches, 649 players)

Metric	Value
statsmodels fit time	11.36s
sklearn fit time	0.08s
Max absolute difference in α	0.033
Median absolute difference in α	0.0002
Pearson correlation of α	0.999996
Max difference in predicted probability	0.001
Winner-prediction agreement (n = 1000)	100.00%

The two methods agree to within four decimal places on α for top players (the largest discrepancies occur for the bottom-ranked players, where sklearn’s tiny L2 penalty exerts marginally more shrinkage), produce identical winner predictions on every sampled match, and yield a Spearman rank correlation of 0.999997. Given equivalent predictions and a $150\times$ speedup, all subsequent forecasts (for Models A, B and C) are completed with Sklearn variants of models. For completeness

of reporting, Model A, B and C as used in previous sections are presented in the relevant section python files. Sklearn variants of models are all presented in `section_5.py`.

Due to the drastically decreased processing time, models were refit daily (it is very unlikely players will play more than one game in day; speed improvements from Sklearn LogReg also allowed for more fits of models). A main *rolling_forecast* function was made that takes in either model inputs, and outputs predicted probabilities and winners for each match in the test set.

5.2. Model Comparison

Problem Statement: Using these forecasts, compare the match-level performance of Models A, B, and C.

Answer:

As expected, accuracy is broadly comparable across models, with Model C showing the largest improvement (61.4% vs 60.9% for A and B). AUC values are similarly close, although Model C also gains a small advantage (0.658 vs 0.652), reflecting that game-level data slightly improves the ranking of predicted strengths.

Brier scores and log-loss show a more nuanced story. Both Models A and B have nearly identical Brier scores (0.238), but Model B has a slightly worse log-loss than Model A (0.748 vs 0.726). This is consistent with the in-sample finding that `home_advantage` does not materially influence the model (section 3.3): adding the home advantage parameter introduces a small amount of noise into predictions without capturing real signal, which log-loss penalises more sensitively than Brier score or accuracy. Model C improves on both Models A and B across all three metrics, with the largest gains in log-loss (0.678) — reflecting that game-level data provides better-calibrated probabilities rather than just better rankings.

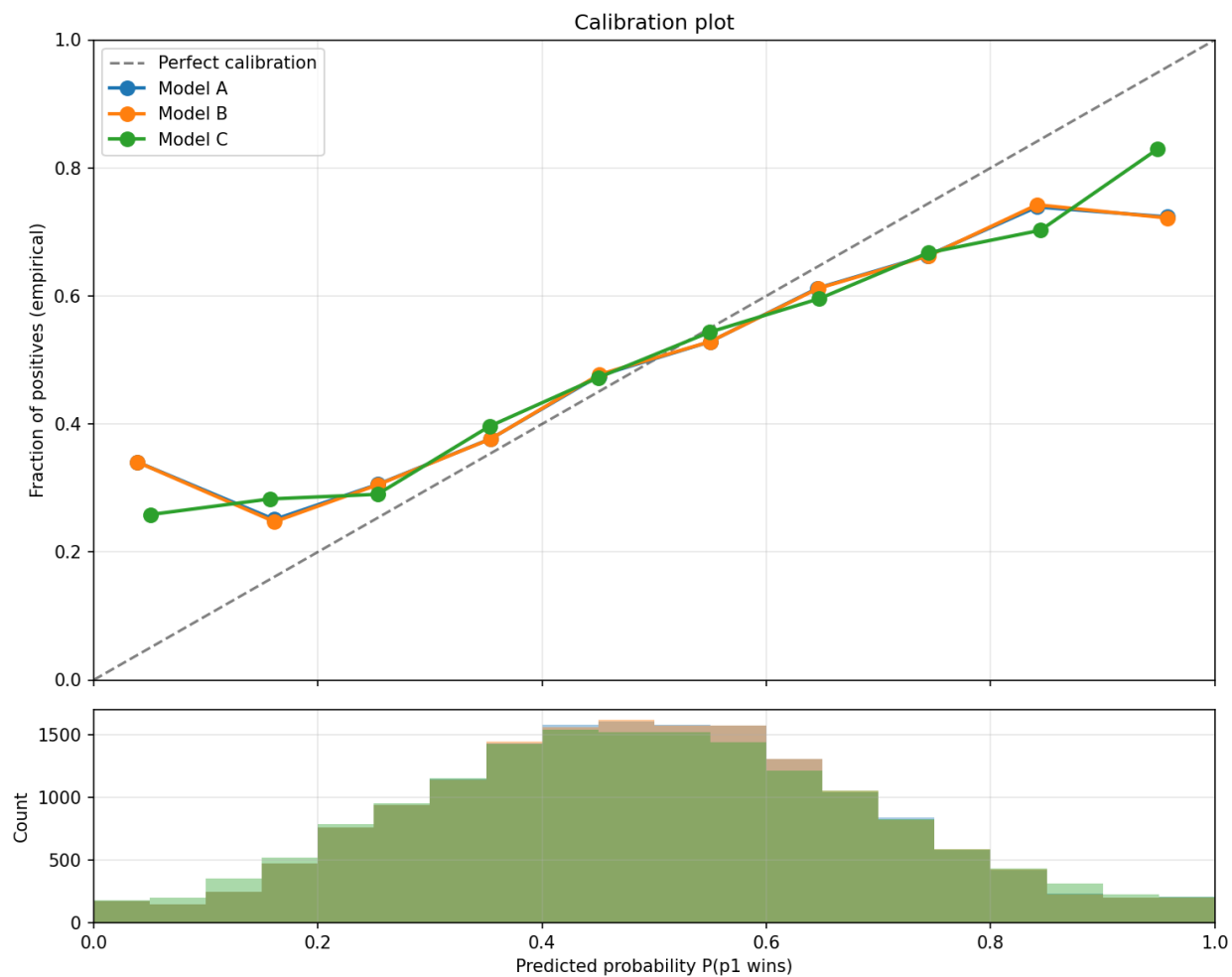
Calibration plots in Figure 3 reveal a similar pattern. All three models track the diagonal closely in the midrange (0.3-0.7), where the bulk of predictions lie, indicating reasonable calibration in this region. The clearest difference appears in the tails, where Model C predicts better calibrated ‘extreme’ probabilities. All models exhibit some overconfidence at the extremes; post-hoc calibration techniques such as temperature scaling could potentially tighten output probabilities towards the ideal line.

Table 8: Forecast Diagnostics for all Models

Model	Log-Loss	Brier Score	Accuracy	AUC-ROC
A	0.7261	0.2380	0.6086	0.6517
B	0.7483	0.2381	0.6085	0.6517
C	0.6776	0.2353	0.6141	0.6580

In summary: Models A and B perform essentially identically, with B marginally worse on log-loss due to the added irrelevant home advantage parameter. Model C provides the most accurate, best-calibrated predictions across all metrics, validating the choice to use richer game-level data when available.

Figure 3: Calibration curves for Models A, B, and C



Appendix

Appendix 1: Descriptive statistics and distribution plots of all matches including outliers

Table A1: Descriptive Statistics by Surface (Untrimmed Dataset)

Surface	Count	Mean	Median	Std	Min	Max
Clay	12,776	9.651	9.500	1.336	6.0	23.667

Surface	Count	Mean	Median	Std	Min	Max
Grass	5,030	10.075	10.000	1.353	6.0	19.800
Hard	16,904	9.817	9.667	1.344	6.0	16.800
Indoor_Hard	6,274	9.925	10.000	1.367	6.0	13.000

Figure A1: Distribution plots showing outliers in untrimmed dataset

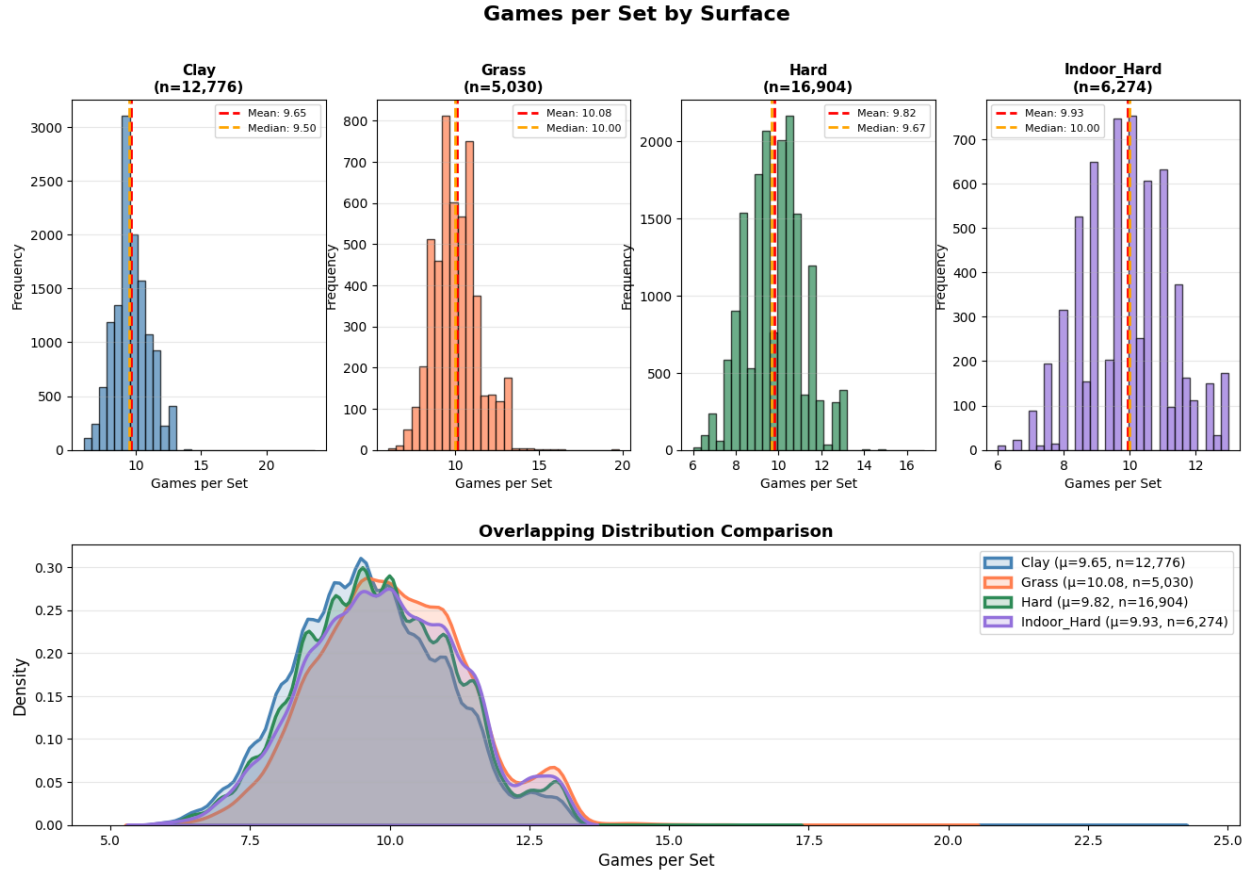


Figure 3: Games per Set by Surface - Untrimmed